

# Demonstration of a Telemetry-Assisted QoS-Aware Scheduler over SOA-Based Time-Slotted Optical Metro-Access Networks

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**Abstract** We demonstrate a telemetry-assisted QoS-aware scheduler controlling an SOA-based time-slotted optical metro-access network. Cyclic plans consider real-time telemetry measuring the state of the queues in the access nodes and the priority of 6G traffic flows. ©2026 The Author(s)

## 1. Review and Demo Use Cases

6G disaggregated radio access networks (RAN) require deterministic fronthaul (FH) transport while supporting hyper reliable and low latency (HRLLC) services [1], [2]. These services typically require multi-access edge computing (MEC) facilities placed at the edge of the network to minimize transmission delay. Emerging 6G use cases such as artificial intelligence (AI) inference and Federated Learning (FL) model transfer are also expected to share the same transport infrastructure [3]. Thus, optical metro-access networks are promising as the first aggregation layer for heterogeneous 6G services over a common infrastructure [3], [4].

Semiconductor optical amplifier (SOA)-based time-slotted optical metro-access networks can support time-sensitive and service-differentiated traffic through nanosecond-scale gating, enabling deterministic forwarding, fine-grained resource sharing, and programmable control [5]. Optical fast switches (OFS)-based node architecture allows intermediate traffic dropping, immediate resource reuse, and direct east-west communication between access sites.

The benefits of such networks can be fully exploited through a quality of service (QoS)-aware scheduler that manages shared time-slotted optical resources across multiple traffic classes [6]. The scheduler prioritizes FH traffic to meet deterministic latency requirements while serving lower-priority flows when resources are available. By using telemetry, the scheduler translates queue states and traffic demand into cycle-based transmission plans and SOA gate-control lists. This demo shows how an east-west communication enabled together with OFS-based optical metro-access network can become an adaptive QoS-aware platform for 6G services.

## 2. Novelty

The novelty of this demo is four-fold: 1) *telemetry-assisted cyclic scheduling* framework for OFS-equipped time-slotted optical metro-access

networks, where queue-state information is translated into cycle-based transmission plans. The scheduler running in a metro location operates asynchronously, while FPGA-based nodes cyclically execute the latest received plan; 2) *hardware-aware execution*, jointly controlling source-node queue selection and SOA gate-control for add/drop/pass operations; 3) *propagation-aware plan activation* using node-dependent cycle offsets; and 4) *priority-aware traffic control* applied to converged 6G traffic, prioritizing FH flows while allocating residual resources for lower-priority flows.

## 3. ECOC Relevance

This demonstration focuses on a highly relevant topic to the ECOC community as it showcases traffic-aware dynamic operation for services of different characteristics, e.g., 6G and AI, which is of the interest of vendors and operators. The programmable and telemetry-assisted time-slotted optical metro-access network unlocks dynamic control, moving beyond static optical transport, which will be received with interest for the audience in general.

## 4. Demo Implementation

**Objective and Configuration:** The objective of this demonstration is to show a telemetry-assisted QoS-aware scheduling workflow over a SOA-based time-slotted optical metro-access network. The demo will show that the proposed cyclic scheduler can satisfy QoS requirements of the traffic to be served over the optical metro-access network.

Fig. 1 shows the schema of the considered testbed and the traffic flows in the uplink direction. At the network level, the testbed focuses on a part of a horseshoe topology that comprises two access (A1, A2) and one metro (M1) location, all equipped with OFS-based nodes. At the 6G RAN, both FH and midhaul (MH) flows are considered. Four Radio Units (RU) (labelled RU1-RU4) support non-HRLLC services, so the generated flows (labelled fh1-fh4) terminate in Distributed Units (DU) placed at the metro location.

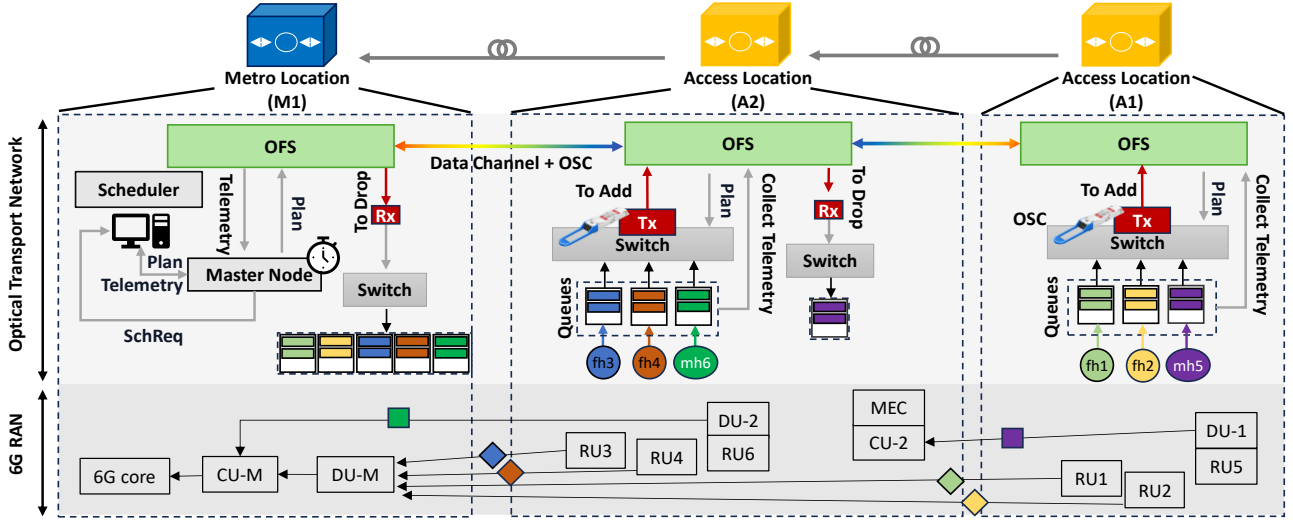


Fig. 1 Schematic of the testbed and examples of traffic flows.

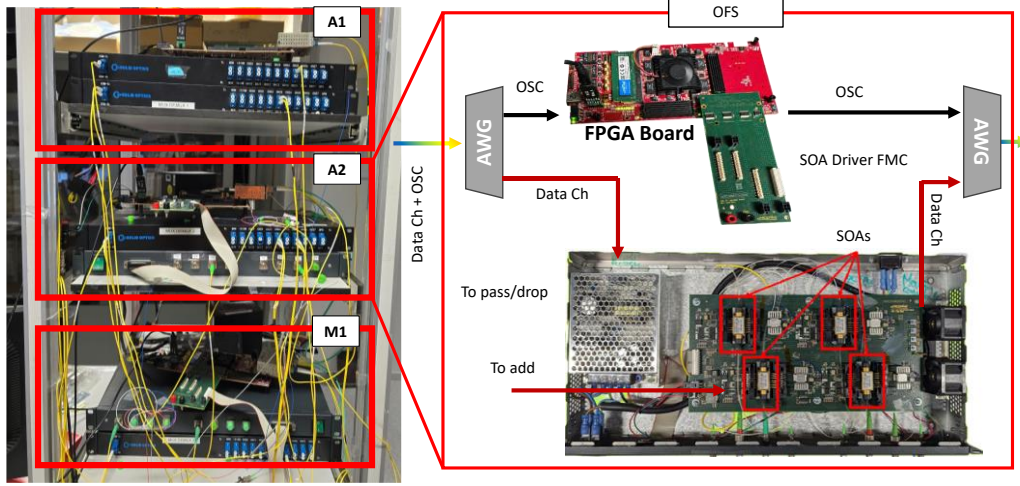


Fig. 2 Physical setup and structure of an OFS within each node.

In contrast, the FH of RUs 5 and 6 terminate at the DUs in A1 and A2, respectively and the MH traffic (mh5 and mh6) is injected to the network. A MEC facility is placed at A1 to support HRLLC services, which connect through RU5.

At the optical data plane, a single wavelength is generated and received by commercial 10G transceivers (0 dBm launch power, 40 km reach). Traffic generators generate FH and MH traffic that are injected to the network. In the control plane, the Optical Supervisory Channel (OSC) transports queue telemetry upstream and distributes scheduling plans downstream, using point-to-point communication with downstream multi-casting and upstream telemetry aggregation.

**Physical Setup and Equipment:** Fig. 2 shows the physical setup for this demonstration. Each node is equipped with an OFS to optically add/drop/pass through the optical signal. The architecture of an OFS is also shown in Fig. 2, where an Arrayed Waveguide Gratings (AWG) separates the OSC and data channel. The OSC is terminated in an FPGA board to read the

scheduling plan and configure the SOAs. To drive the SOAs, we employ custom-made SOA drivers controlled via the FPGA board's FMC (FPGA Mezzanine Card) interfaces [5]. The FPGA at each OFS is also responsible for collecting telemetry from the local queues and generating telemetry reports.

Each access node includes an electrical switch that selects packets from queues according to the scheduling plan, controlled by the FPGA. At the metro location (see Fig. 1), the master node periodically (with period  $T_s$ ) generates scheduling requests (SchReq), forwarding them to the scheduler module. Once computation completes (it takes a few hundred milliseconds), the plan is distributed over the OSC; meanwhile, FPGAs continue executing the last received plan. Each plan covers one cycle of duration ( $T_c \ll T_s$ ) and is repeated over consecutive cycles. Note that the plan must be computed to absorb variations of the traffic flows during long scheduling cycles without impacting the QoS of the flows.

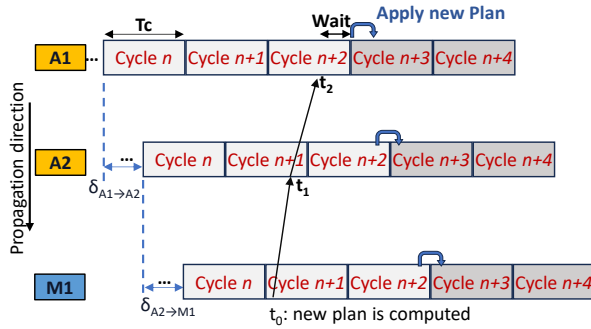


Fig. 3 Plan activation mechanism.

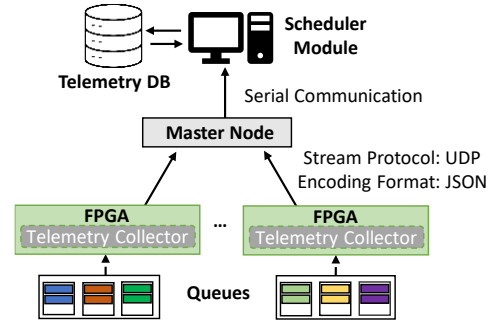


Fig. 4 Telemetry collection procedure.

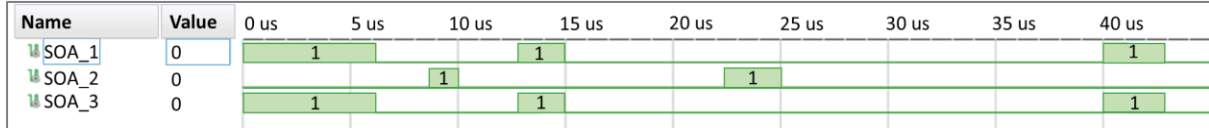


Fig. 5 Vivado lab dashboard to show the status of the SOAs.

Fig. 3 illustrates the plan activation mechanism. To avoid explicitly accounting for propagation delay, each node applies a predefined time offset equal to the cumulative propagation latency from the upstream node. For instance, node A2 starts its local cycle after an offset  $\delta_{A1 \rightarrow A2}$  relative to A1, while M1 starts after  $\delta_{A1 \rightarrow A2} + \delta_{A2 \rightarrow M1}$ . When a new plan is received, A1 waits until the next cycle boundary to apply it, and each downstream node (synchronized and aware of its offset) activates the plan at the start of its corresponding local cycle, ensuring collision-free execution. The plan specifies each flow's starting timeslot, transmission duration, and SOA gate configuration. Once distributed, each access node's FPGA applies it to the SOA gates and queue selection switch.

As shown in Fig. 4, each FPGA includes a telemetry collector that reads local queue information and generates reports containing input bytes and queued bytes remaining after service, with granularity equal to the cycle duration  $T_c$ . To limit OSC overhead, telemetry is collected every  $T_{col}$  (an integer multiple of  $T_c$ ), encoded in JSON, and streamed via UDP to the master node, which forwards it to the scheduler module via serial communication. The scheduler stores telemetry measurements in a database (DB) and uses them for plan computation, applying a safety margin to account for traffic variations.

**Presentation to Attendees:** Since the experimental testbed consists of fixed optical components, FPGA boards, SOA drivers, and measurement equipment, it will remain deployed in the laboratory at TU/e premises. For the demonstration, a remote connection will be established to a control PC located in the lab, which is responsible for computing the scheduling plan and communicating with FPGAs, allowing the live operation of the metro-access network.

Attendees will observe the complete

telemetry-assisted scheduling workflow, including collected telemetry samples, generated scheduling plans, SOA gate-control traces at the nodes, and measured latency. The scheduler output will be shown together with representative telemetry reports and the corresponding plan applied by the FPGA-based nodes.

**Attendee Interaction:** Attendees will be able to interact with the demonstration in two ways. First, they will be able to select which SOA trace should be displayed using the Vivado Lab Edition dashboard [7] (see Fig. 5) and verify its compatibility with the computed plan. This interaction allows attendees to directly follow the relation between queue telemetry, plan computation, FPGA cyclic plan execution, and SOA gate operation.

Second, a graphical user interface (GUI) dashboard will display latency plots for all configured flows; the audience will be able to select among the flows to inspect latency performance under the QoS-aware scheduling policy. In case of connection loss or any issue with the remote access to the laboratory setup, a pre-recorded video of the full workflow will be available to ensure that the complete demonstration can still be presented.

## 5. Conclusion

This demonstration will showcase a telemetry-assisted scheduler for SOA-based time-slotted optical metro-access networks targeting service-differentiated 6G transport. By exploiting queue telemetry, the scheduler adapts cyclic plans to the current traffic state, prioritizing latency-sensitive flows while reusing residual resources for lower-priority services. The demo highlights how FPGA-based execution, SOA gate control, and propagation-aware plan activation can be jointly realized, offering a practical path toward adaptive, QoS-aware optical metro-access networks for deterministic 6G transport.

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